

Network Inventory Analysis

This report documents the current status and configuration of the 15-device network infrastructure, comprising core routers, access switches, servers, and end-user devices.

Device Inventory Summary

Type	Count	Key Details
Core Routers	3	ISR4331 (WAN Hub), 2911s (Gateways)
Switches	4	2960-24TT (L2 Access)
Servers	4	Server-PT (Services/Data)
PCs	4	End-user endpoints

Core Routers

Router0: WAN Hub (ISR4331)
Router6: Site A (2911)
Router7: Site B (2911)

Infrastructure

4x Layer 2 Switches (2960-24TT)
4x DHCP-enabled Servers
4x User PCs

Gateway Configuration Analysis

Router0: WAN Hub (ISR4331)

The centralized hub connecting Site A and B. Implements restrictive ingress/egress filtering for security.

- **Interfaces:** Gi0/0/0 (Site A), Gi0/0/1 (Site B)
- **Routing:** RIP (Propagating WAN subnets)
- **Security:** Implicit DENY (RFC1918 spoofing check)

Router6: Site A Gateway

Manages LAN subnets (VLAN 10/20) and enforces traffic policy.

- **Interfaces:** WAN to Router0, VLAN 10 (192.168.1.0), VLAN 20 (192.168.2.0)
- **Security ACL (Outbound Gi0/0):** Restrictive filtering. Blocks VLAN 10 outbound traffic.

Router7: Site B Gateway

Manages Site B server subnets (VLAN 30/40).

- **Interfaces:** WAN to Router0, VLAN 30 (202.20.20.0), VLAN 40 (201.10.10.0)
- **Routing:** RIP (Standard dynamic routing)

Data Flow & Connectivity

Intra-Site Connectivity

Source	Destination	Path	Status
PC3	Server1	Switch1 (L2)	✓
PC5	Server2	Switch2 (L2)	✓
PC3	PC5	Router6	✓

Inter-Site Connectivity

Connectivity between sites relies on RIP-advertised routes, but is subject to ACL filtering.

- **PC3 (Site A) to Server3 (Site B):** Successful (via Router0/7)
- **Server1 (Site A) to Server4 (Site B):** ⚠ Filtered (ACL 10 on Router6 blocks 192.168.1.0)
- **PC5 (Site A) to Server4 (Site B):** Successful (VLAN 20 permitted)

Configuration Summary Matrix

Aspect	Router0	Router6	Router7
Model	ISR4331	2911	2911
WAN IPs	200.1.2.1, 200.2.3.2	200.1.2.2	200.2.3.1
LAN Usage	None	VLAN 10/20	VLAN 30/40
ACL Status	None	ACL 10 (Out)	None

Baseline Switch Configuration

All switches (Switch1-4) leverage identical baseline configurations focused on defense-in-depth:

- **Security:** Firewall integration, RFC1918 spoofing protection.
- **Separation:** VLAN isolation (Web vs. Database vs. Security).
- **Status:** Spanning tree enabled; no VLAN trunking; all ports access-mode.

Design Assessment

Strengths

- **Dynamic Routing:** RIP automates table updates.
- **Security Policy:** Enforces access list concepts at the routing layer.
- **Service Architecture:** DHCP servers pre-configured for each subnet.

Weaknesses

- **Topology:** Single WAN path creates a critical point of failure.
- **Asymmetry:** Confusing ACLs block traffic unevenly.
- **Protocol:** RIP is outdated with long convergence times.
- **Layer 2/3:** Lack of VLAN trunking and SVIs limits scaling.

Recommendations

To improve resilience and scalability:

1. Upgrade RIP to OSPF for faster convergence and better scalability.
2. Implement redundant WAN links to eliminate the Router0 failure point.
3. Deploy Inter-VLAN routing (SVIs) on distribution layer switches.
4. Rationalize ACL policies to ensure symmetric traffic flow between segments.